

Customer360 Data Warehouse – Data Quality and ETL Report

A. Data Source

The data used for this project comes from the original Customer360 raw data file. The file contains information about customers, accounts, products, transactions, and CRM interactions. The source data includes fields such as customer details, product types, account information, transaction amounts, interaction types, and dates.

The raw file is the starting point of the data warehouse process. The data is first collected from this source before being loaded into the staging area.

B. Data Staging

The raw data is loaded into a staging table before it is moved into the data warehouse. The staging area acts as a temporary storage location where the original source data can be kept.

The staging table was created to match the information provided in the raw file. The data in the staging area was not cleaned or changed because the purpose of staging is to preserve the original source data.

Keeping the raw data unchanged also makes it easier to identify and investigate data-quality problems during the ETL process.

C. Data Warehouse Design – Star Schema

The data was organised using a star schema. The star schema separates descriptive information into dimension tables and business activities into fact tables.

The dimension tables contain information such as customer, product, date, and other descriptive information. The fact tables contain business events and measurable information such as transactions and CRM interactions.

The use of a star schema makes the data easier to query and analyse. It also allows reports to combine information from different parts of the Customer360 dataset.

D. Data Transformation and ETL Process

The ETL process is used to move data from the staging area into the final data warehouse.

During this process, the data is checked and transformed before it is loaded into the dimension and fact tables. Duplicate records, missing values, and inconsistent information are considered as part of the data-quality checks.

Surrogate keys are used in the dimension tables to provide unique identifiers for warehouse records. The appropriate information is then linked between the dimension and fact tables.

The staging data is kept unchanged, while the necessary transformations and data-quality handling are performed during the ETL process.

E. Final Data Warehouse Layer

After the ETL process, the transformed data is stored in the Customer360 data warehouse.

The final warehouse contains the dimension and fact tables that form the star schema. This provides a structured and consistent environment for storing historical business information and supporting analysis.

The final data warehouse can then be used by SQL queries, views, and reporting tools without having to work directly with the original raw file.

F. Reporting and Data Analysis

SQL queries and views are used to analyse the information stored in the data warehouse.

The reports can answer business questions such as:

- How many customers are in each province?
- How many customers hold each product type?
- How many customers hold more than one product?
- What is the total and average account balance by product type?
- Which customers have a Savings account but no Credit Card?
- What percentage of Credit Card accounts are close to their credit limit?
- What is the relationship between CRM interactions and transaction value?
- Which accounts show unusual transaction patterns?

The reporting layer allows the information stored in the data warehouse to be presented in a useful format for analysis and decision-making.

G. Data Quality Assessment and Handling

The raw file was reviewed for common data-quality issues such as duplicate customer records, missing values, inconsistent values, and incomplete information.

The staging table was kept unchanged so that the original source data could be preserved. This means that the staging layer represents the data as it was received.

During the ETL process, data-quality issues were handled before the data was loaded into the final warehouse tables. Duplicate records were identified where necessary, while missing or unknown information was not replaced with made-up values. Where a value could not be reliably determined, it was kept as NULL or represented as unknown.

This approach helps maintain the integrity of the original data while ensuring that the final data warehouse is structured and suitable for reporting and analysis.